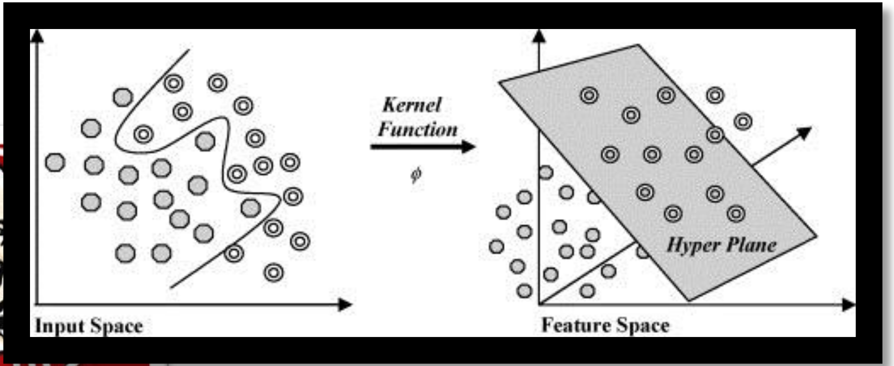
KERNELS



INTUITION

- Many classifiers are not linear, but have some sort of pattern, such as a circle or a curve.
- The intuition behind kernel methods is to find a map that maps the current space into a higher dimensional space, so that it is easier to learn the classifier in the higher dimensional space.
- This map is called a feature map.

FEATURE MAPS

Example. Non-linear classifiers.

$$x = (x_1, x_2)$$

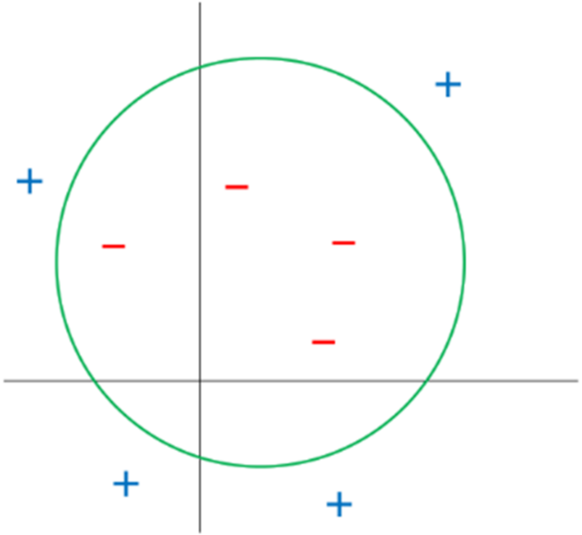
$$\phi(x) = \left(1, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}\, x_1 x_2, x_1^2, x_2^2\right)$$

$$\begin{aligned} h(x; \theta, \theta_0) & \\ &= \operatorname{sign}(\theta \cdot \phi(x)) \\ &= \operatorname{sign}\left(\theta_1 + \theta_2\sqrt{2}x_1 + \theta_3\sqrt{2}x_2 + \theta_4\sqrt{2}x_1 x_2 + \theta_5 x_1^2 + \theta_6 x_2^2\right) \end{aligned}$$

FEATURE MAPS

Example. Non-linear classifiers.

$$h(x;\theta,\theta_0)$$
$$= \mathrm{sign}\big((x_1-1)^2+(x_2-2)^2-9\big)$$
$$= \mathrm{sign}\big(-4-2x_1-4x_2+x_1^2+x_2^2\big)$$



COMPUTATIONAL CHALLENGE

High-Dimensional Features.

$$x = (x_1, x_2, \dots, x_{1000}) \in \mathbb{R}^{1000}$$

$$\phi(x) = \left(1, \dots, \sqrt{2}x_i, \dots, \sqrt{2}x_i x_j, \dots, x_i^2, \dots\right) \in \mathbb{R}^{501501}$$

Inner Products.

Computing $\phi(x) \cdot \phi(x')$ for $x, x' \in \mathbb{R}^{1000}$ requires about 2,004,000 floating point operations.

CC: SOLUTION

- The higher the dimension of the feature map, the more complex the computation.
- What if we can find feature maps such that certain functions of the variables in the higher dimensional space can be computed by variables in the original lower dimensional space?
- E.g. DOT PRODUCT
- If working out the classifier only requires us to know only these functions, then we save on computational complexity!

DOT/INNER PRODUCT

Fortunately, many inner products simplify nicely.

$$
\begin{aligned}
K(x, x') &= \phi(x) \cdot \phi(x') \\
&= 1 + 2\textstyle\sum_i x_i x_i' + 2\sum_{i<j} x_i x_j x_i' x_j' + \sum_i x_i^2 x_i'^2 \\
&= 1 + 2(\textstyle\sum_i x_i x_i') + (\sum_i x_i x_i')^2 \\
&= (x \cdot x' + 1)^2
\end{aligned}
$$

For $x, x' \in \mathbb{R}^{1000}$, computing this requires only about 2000 floating point operations, less than the 501,501 operations needed for $\phi(x)$.

KERNEL TRICK

The kernel trick refers to the strategy of converting a learning algorithm and the resulting predictor into ones that involve only the computation of the kernel $K(x, x') = \phi(x) \cdot \phi(x')$ but not of the feature map $\phi(x)$.

WHAT IS A KERNEL

'Infinite dimensional' positive definite matrices

Definition.

A function $K: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a *kernel function* if

1. $K(x,y) = K(y,x)$ for all $x, y \in \mathbb{R}^d$,
2. given $n \in \mathbb{N}$ and $x^{(1)}, x^{(2)}, \dots, x^{(n)} \in \mathbb{R}^d$,

   the *Gram matrix* K with entries

   $$K_{ij} = K\left(x^{(i)}, x^{(j)}\right)$$

   is positive semidefinite.

Example. $K(x, x') = \phi(x) \cdot \phi(x')$

Can be shown that all kernel functions are of this form!

EXAMPLES

Linear Kernel.

$$K(x, x') = x \cdot x'$$

Polynomial Kernel.

$$K(x, x') = (x \cdot x' + 1)^k$$

Radial Basis Kernel.

$$K(x, x') = \exp\left(-\frac{1}{2}\|x - x'\|^2\right)$$

Feature map $\phi(x)$ is infinite dimensional!

RIDGE REGRESSION

Recall for linear regression with L2 regularization, we aim to minimize the loss function

$$\mathcal{L}(\theta) = \frac{1}{2}\sum_{i=1}^{m}\left(\left\langle \theta, \phi\left(x^{(i)}\right)\right\rangle - y^{(i)}\right)^2 + \frac{\lambda}{2}\|\theta\|^2 .$$

Taking the gradient and setting to zero, we have

$$\begin{aligned}\theta &= -\frac{1}{\lambda}\sum_{i=1}^{m}\left(\left\langle \theta, \phi\left(x^{(i)}\right)\right\rangle - y^{(i)}\right)\phi\left(x^{(i)}\right)\\ &= \sum_{i=1}^{m} a_i \phi\left(x^{(i)}\right),\end{aligned}$$

where $a_i = -\frac{1}{\lambda}\left(\left\langle \theta, \phi\left(x^{(i)}\right)\right\rangle - y^{(i)}\right)$.

RIDGE REGRESSION

We can write the expression above as $\mathbf{\Phi}^T \mathbf{a}$, where $\mathbf{\Phi}$ is the design matrix whose i^{th} row is given by $\phi\left(x^{(i)}\right)^T$, and where $a = (a_i, \ldots, a_m)^T$. Substituting $\theta = \mathbf{\Phi}^T a$ into the loss function, we have

$$\mathcal{L}(a) = \frac{1}{2}\left\langle \mathbf{\Phi\Phi}^T a,\, \mathbf{\Phi\Phi}^T a \right\rangle - \left\langle \mathbf{\Phi\Phi}^T a,\, y \right\rangle + \frac{1}{2}\|y\|^2 + \frac{\lambda}{2}\left\langle a, \mathbf{\Phi\Phi}^T a \right\rangle,$$

where $y = \left(y^{(1)}, \ldots, y^{(m)}\right)^T$.

RIDGE REGRESSION

Now define $K = \mathbf{\Phi}\mathbf{\Phi}^T$, where the ij^{th} element of K is given by

$$\left\langle \phi\left(x^{(i)}\right), \phi\left(x^{(j)}\right)\right\rangle = k\left(x^{(i)}, x^{(j)}\right),$$

and we have

$$\mathcal{L}(a) = \frac{1}{2}\left\langle a, K^2 a\right\rangle - \langle Ka, y\rangle + \frac{1}{2}\|y\|^2 + \frac{\lambda}{2}\langle a, Ka\rangle .$$

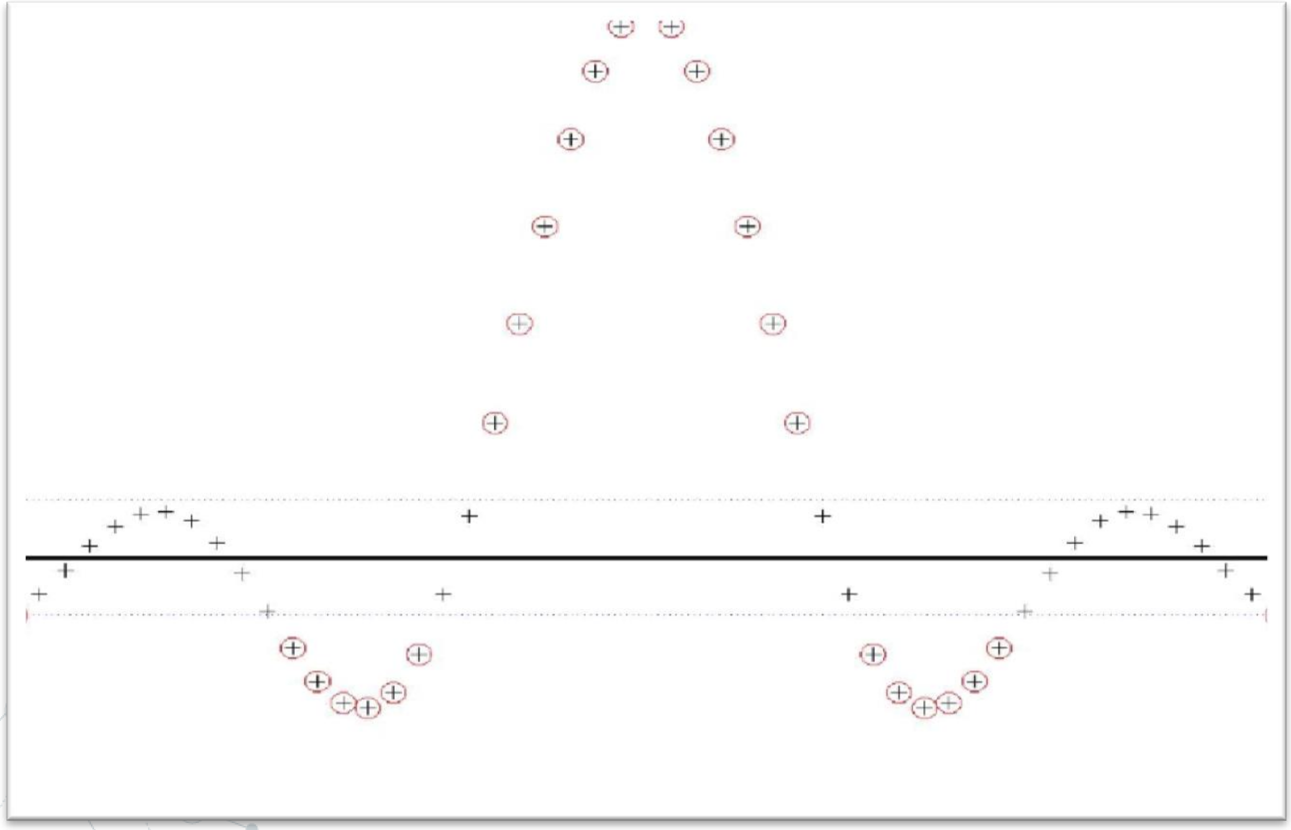
RIDGE REGRESSION

Taking the gradient with respect to a and setting it to zero, we obtain

$$a = (K + \lambda I)^{-1} y,$$

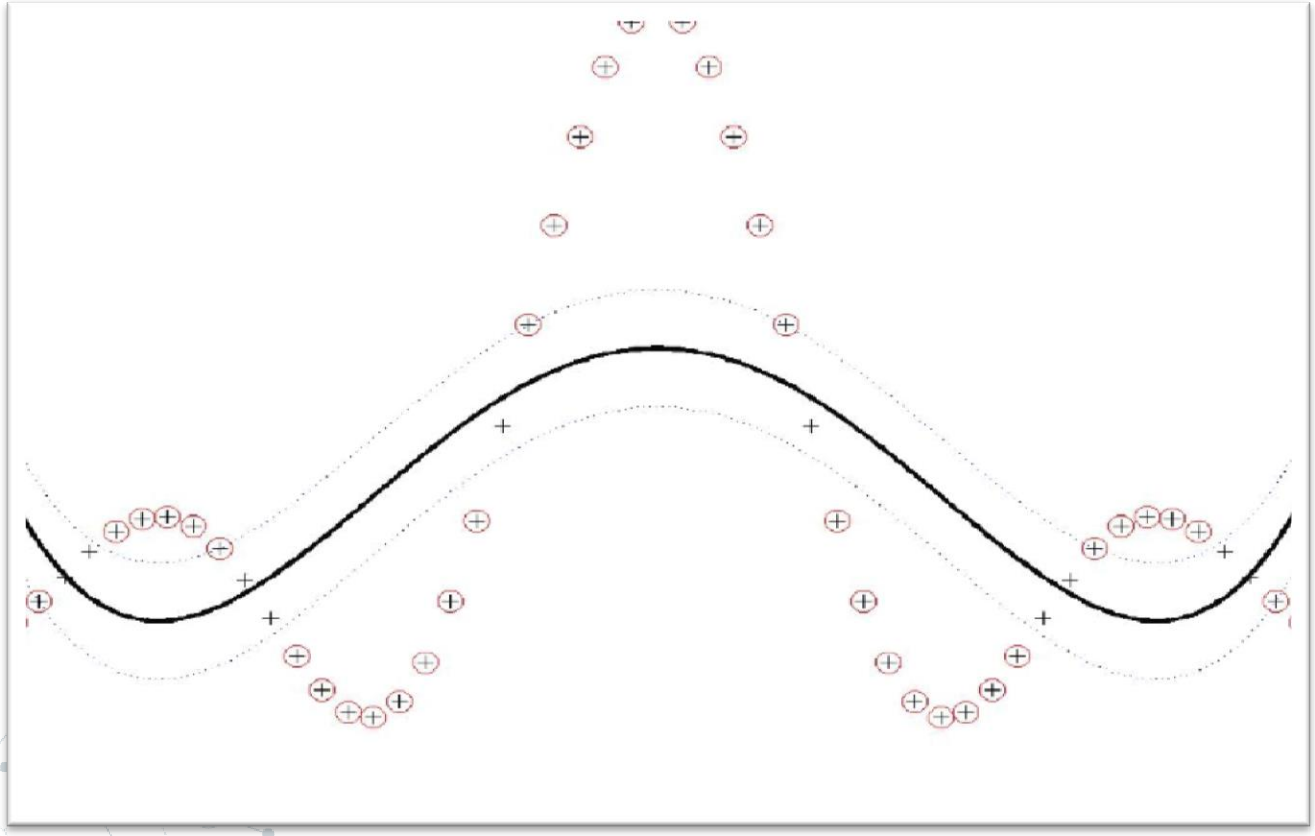
and we see that we can express the solution entirely in terms of the kernel.

EXAMPLE



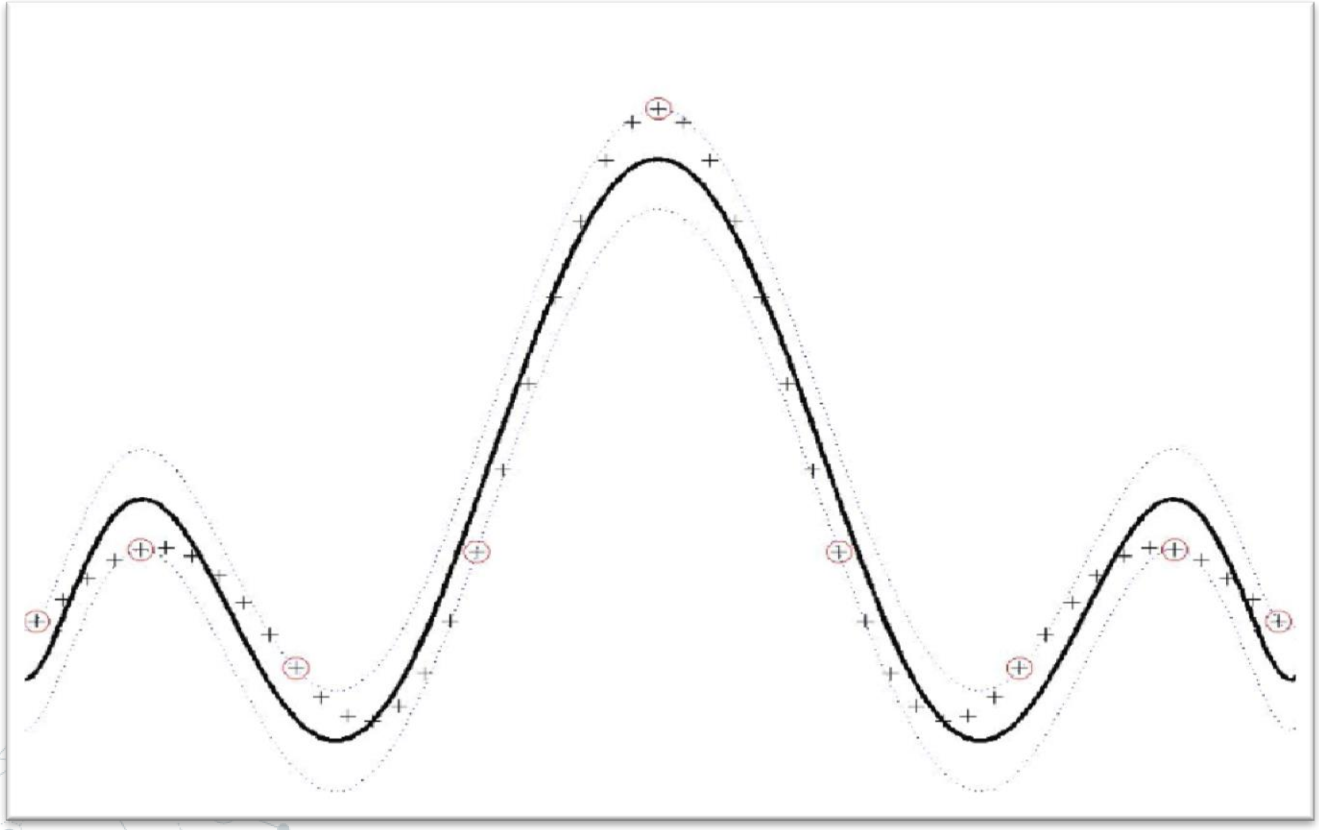
1st order
Polynomial
Kernel

EXAMPLE



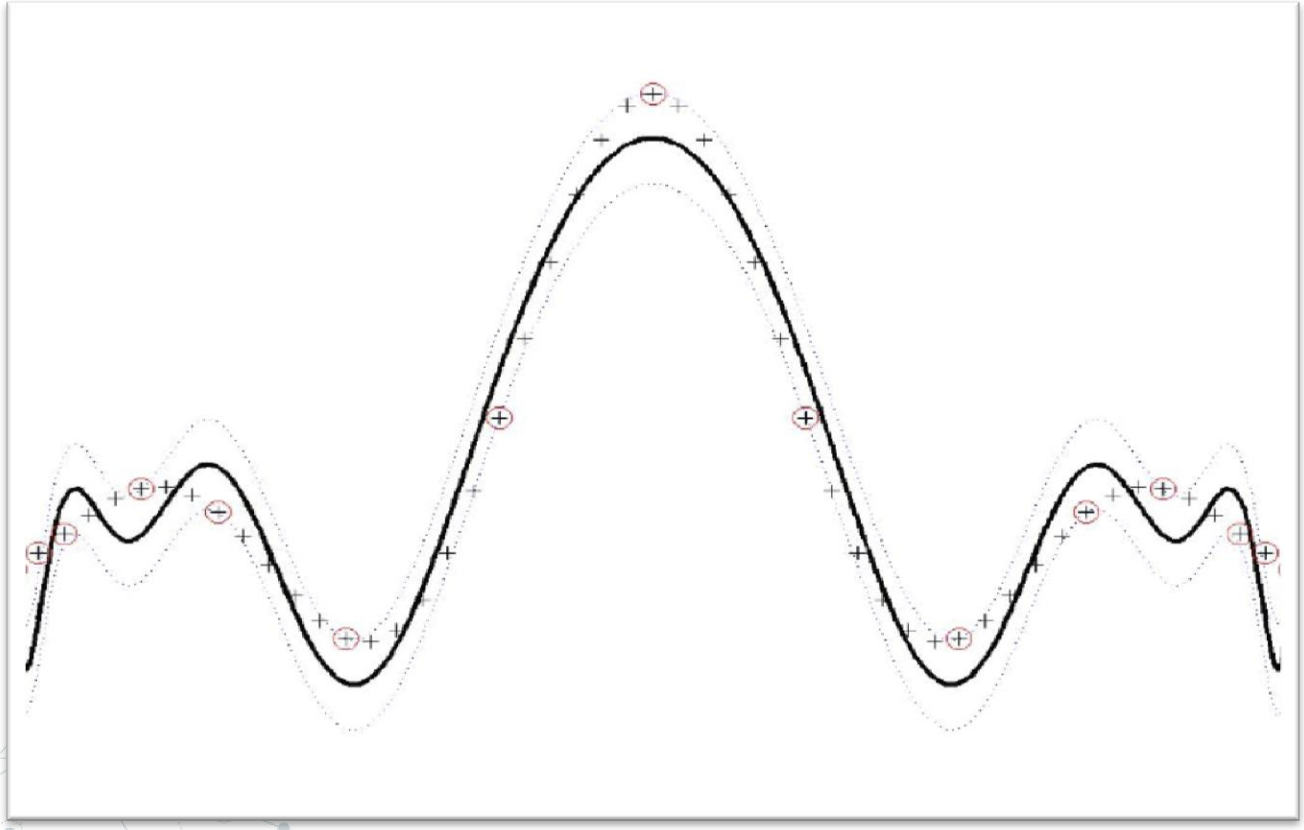
4^{th} order Polynomial Kernel

EXAMPLE



13^{th} order Polynomial Kernel

EXAMPLE



33^{rd} order Polynomial Kernel

Theorem (Mercer)

Let $\mathcal{X}$ be a closed and bounded subset of $\mathbb{R}^d$ and μ be a finite measure on $\mathcal{X}$. Let $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a continuous symmetric positive-definite kernel on $\mathcal{X}$. Define the operator

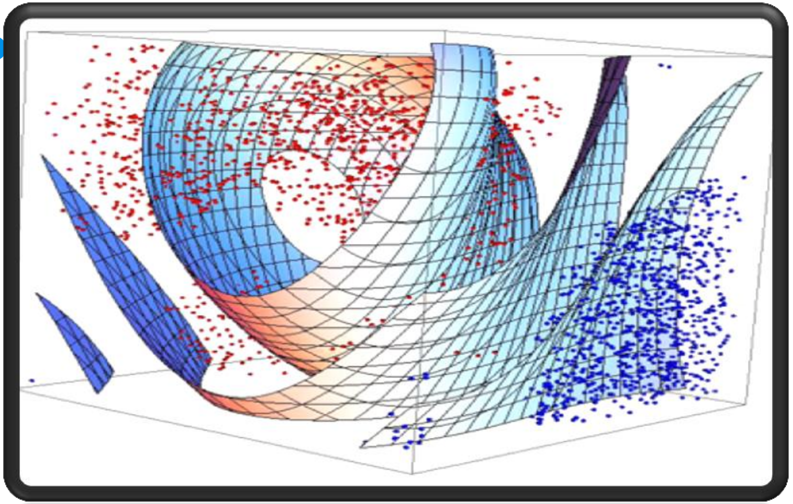
$$T_K f(x) = \int_{\mathcal{X}} K(x,t) f(t) \,\mathrm{d}\mu(t)$$

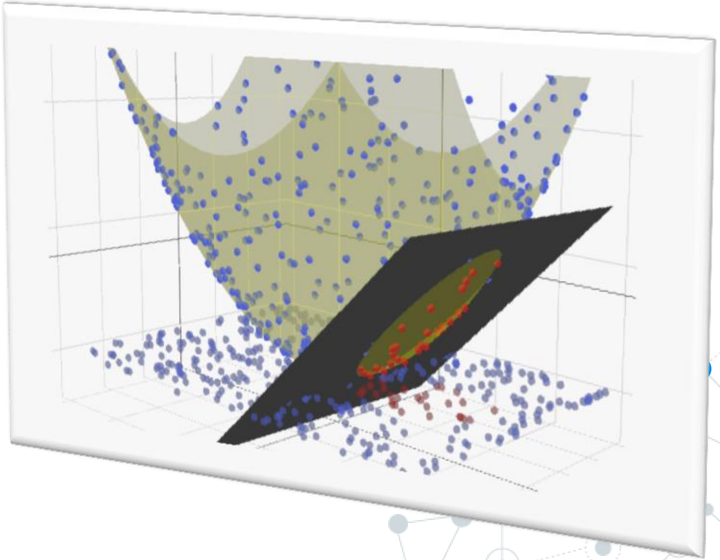
for all $f \in L^2(\mathcal{X}, \mu)$. Then there exists an orthonormal basis $\{\phi_i\}$ of eigenfunctions of T_K, with corresponding non-negative eigenvalues λ_i, such that

$$K(x,y) = \sum_{i=1}^{\infty} \lambda_i \phi_i(x) \phi_i(y)$$

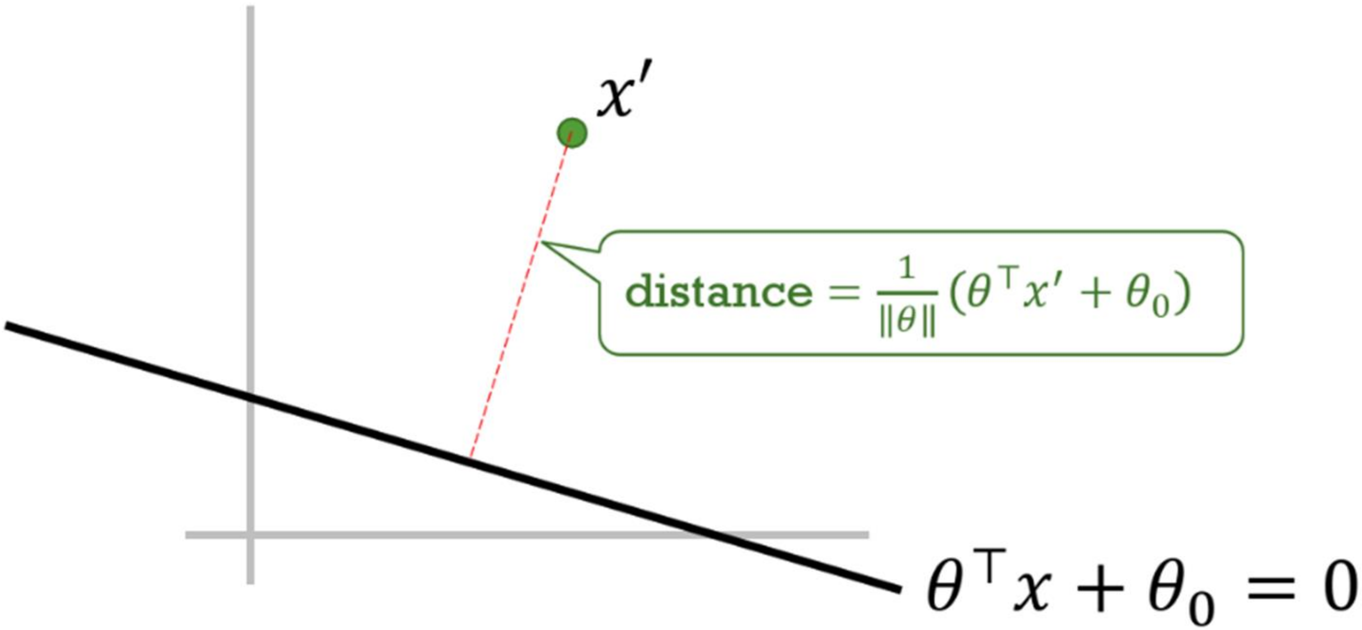
where the convergence is absolute and uniform.

SUPPORT VECTOR MACHINES





MAXIMUM MARGINS



MAXIMUM MARGINS

Unfortunately, this only applies to data that is linearly separable.

Our goal is to

maximize $1/\|\theta\|$

subject to $y(\theta^\top x + \theta_0) \geq 1$ for all data (x, y)

Or equivalently,

minimize $\frac{1}{2}\|\theta\|^2$

subject to $y(\theta^\top x + \theta_0) \geq 1$ for all data (x, y)

MAXIMUM MARGINS

Drop θ_0 for now

minimize $\frac{1}{2}\|\theta\|^2$

subject to $y(\theta^\top x) \geq 1$ for all data (x, y)

Lagrangian. $L(\theta, \alpha) = \frac{1}{2}\|\theta\|^2 + \sum_{(x,y)} \alpha_{x,y} \left(1 - y(\theta^\top x)\right)$

To find $\ell(\alpha) = \min_\theta L(\theta, \alpha)$, we solve

$$0 = \nabla_\theta L(\theta, \alpha) = \theta - \sum_{(x,y)} \alpha_{x,y}\, yx$$

to get $\theta = \sum_{(x,y)} \alpha_{x,y}\, yx$. Substituting into $L(\theta, \alpha)$ gives

$$\ell(\alpha) = \sum_{(x,y)} \alpha_{x,y} - \frac{1}{2} \sum_{(x,y)} \sum_{(x',y')} \alpha_{x,y}\, \alpha_{x',y'} yy'(x^\top x').$$

MAXIMUM MARGINS

It can be shown that the primal and dual problems are equivalent (*strong duality*).

Primal.

minimize $\frac{1}{2}\|\theta\|^2$

subject to $y(\theta^\top x) \geq 1$ for all data (x, y)

Dual.

maximize $\sum_{(x,y)} \alpha_{x,y} - \frac{1}{2}\sum_{(x,y)}\sum_{(x',y')} \alpha_{x,y}\,\alpha_{x',y'}yy'(x^\top x')$

subject to $\alpha_{x,y} \geq 0$ for all (x, y)

After solving the dual to get the optimal $\alpha_{x,y}$'s, we obtain the optimal θ using $\theta = \sum_{(x,y)} \alpha_{x,y}\, yx$.

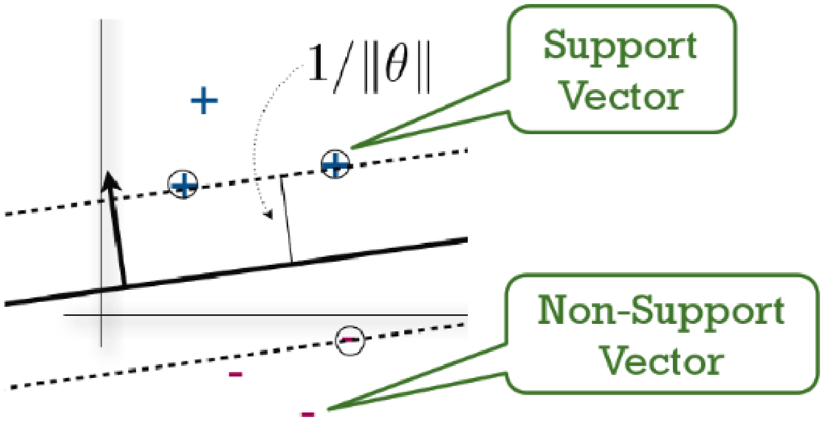
SUPPORT VECTORS

Complementary Slackness.

$\hat{\alpha}_{x,y} > 0$:	$y(\hat{\theta}^\top x) = 1$	Support Vectors
$\hat{\alpha}_{x,y} = 0$:	$y(\hat{\theta}^\top x) > 1$	Non-Support Vectors

Sparsity

Since very few data points are support vectors, most of the $\hat{\alpha}_{x,y}$ will be zero.



KERNEL TRICK

Learning.

$$\ell(\alpha) = \sum_{(x,y)} \alpha_{x,y} - \frac{1}{2}\sum_{(x,y)}\sum_{(x',y')} \alpha_{x,y}\,\alpha_{x',y'}\,yy'(x^\top x')$$

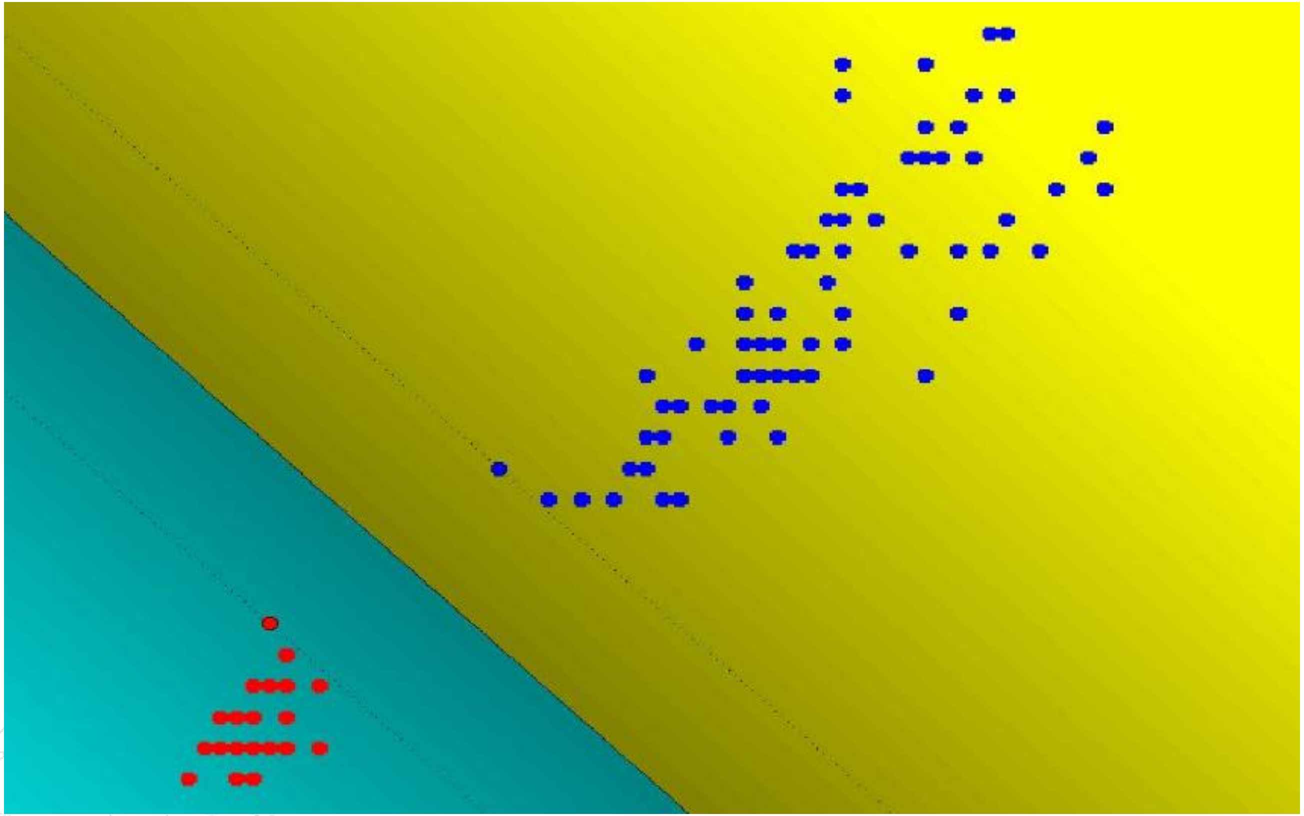
Prediction.

$$h(x;\theta) = \mathrm{sign}(\theta^\top x) = \mathrm{sign}\left(\sum_{(x',y')} \alpha_{x',y'}\,y'(x^\top x')\right)$$

For the dual, we don't need the feature vectors x, x'.
Knowing just the dot products $(x^\top x')$ is enough.

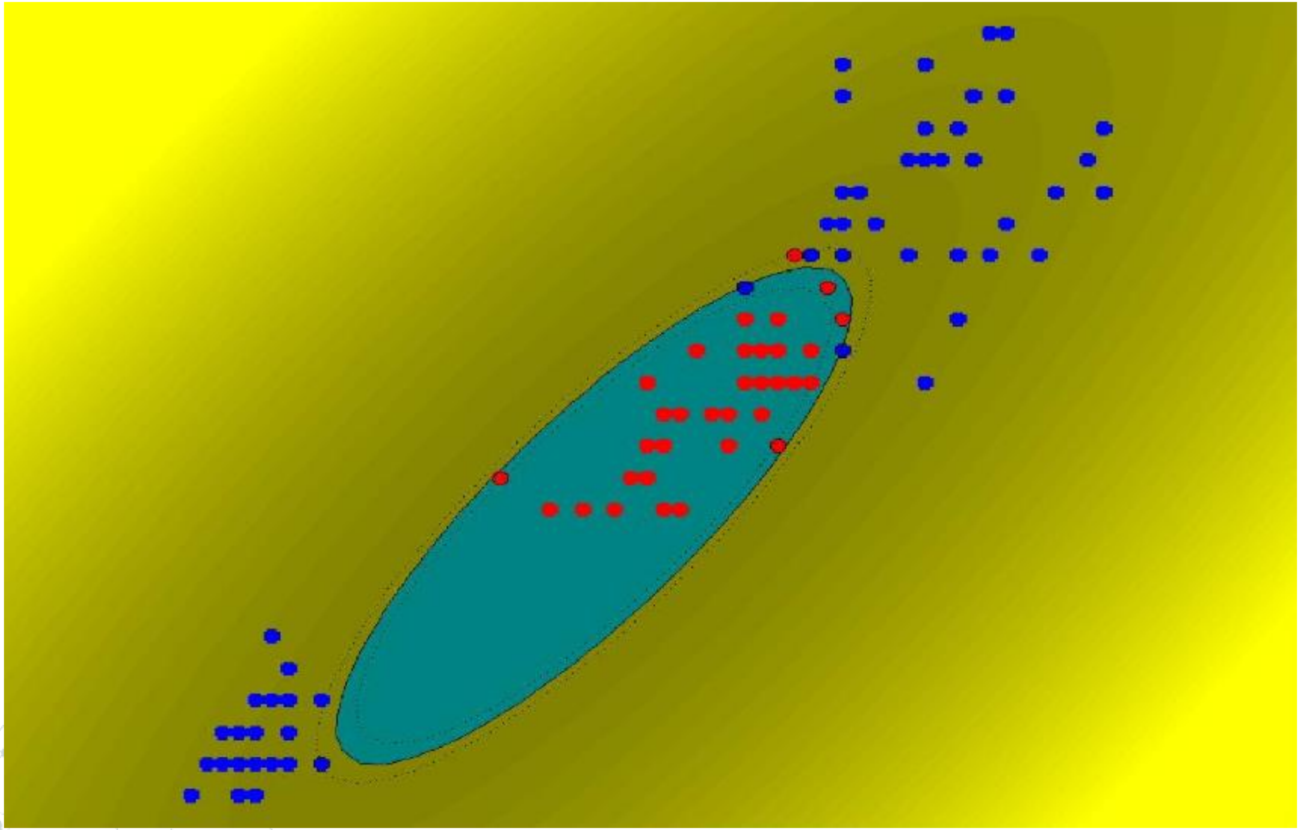
Recall that $(x^\top x')$ is a measure of similarity between x and x'.
This similarity function is also called a *kernel*.

IRIS DATA SET 1 VS 2,3



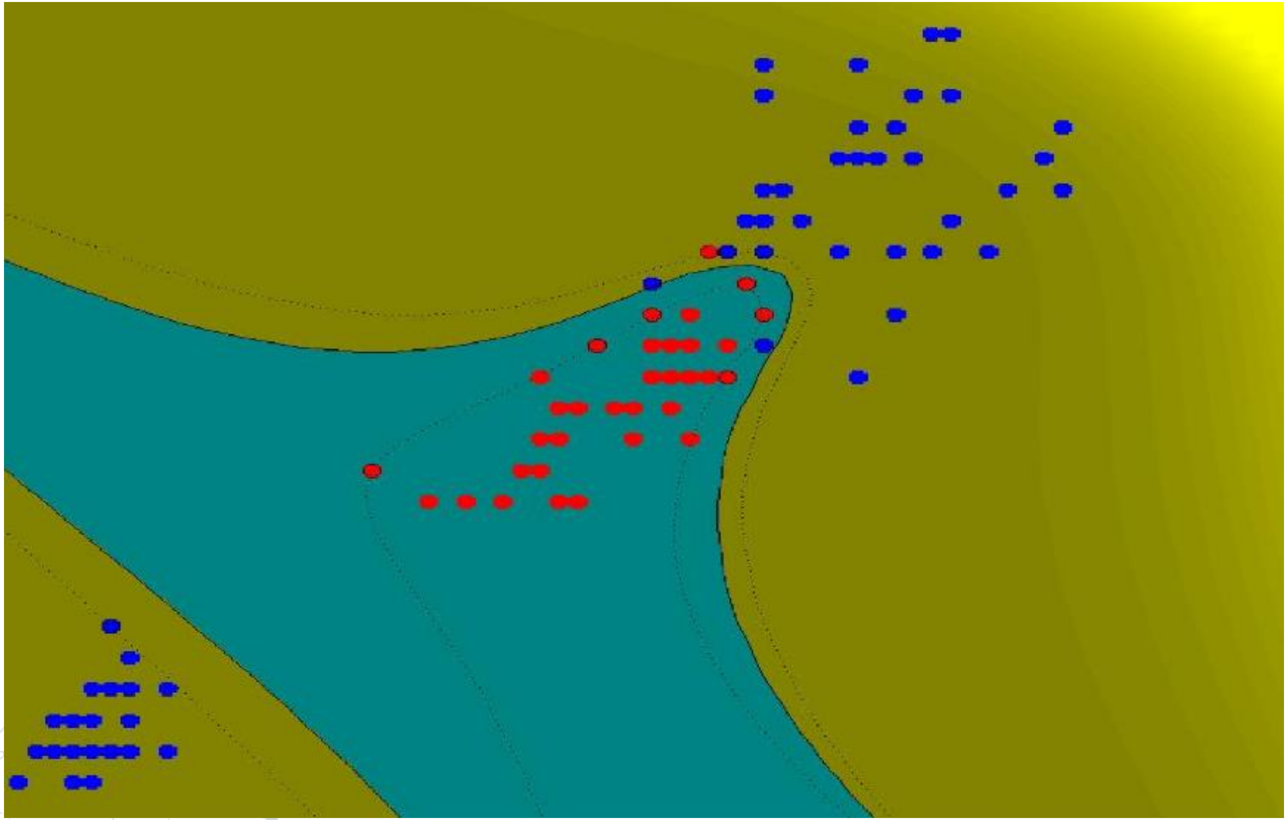
Linear
Kernel

IRIS DATA SET 2 VS 1,3



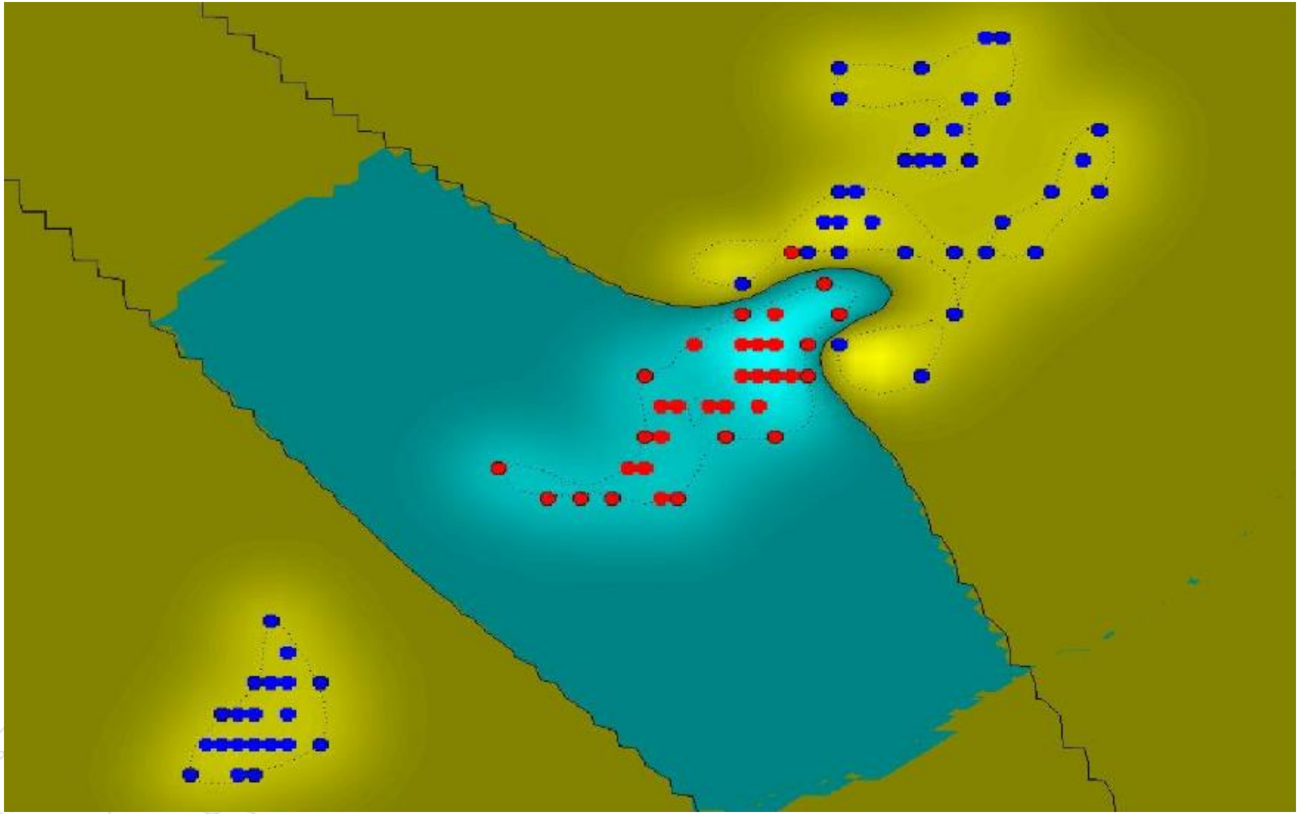
2nd order Polynomial Kernel

IRIS DATA SET 2 VS 1,3



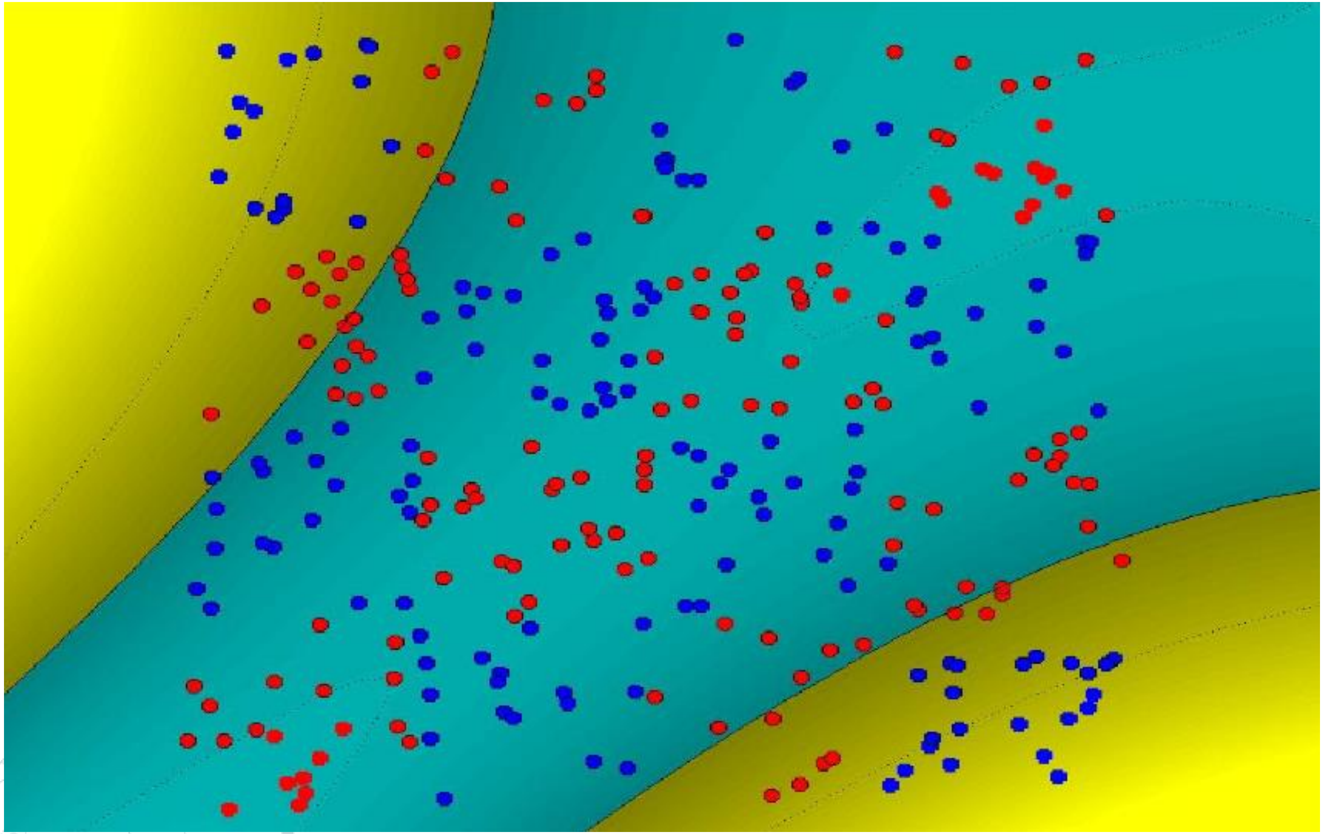
13th order Polynomial Kernel

IRIS DATA SET 2 VS 1,3



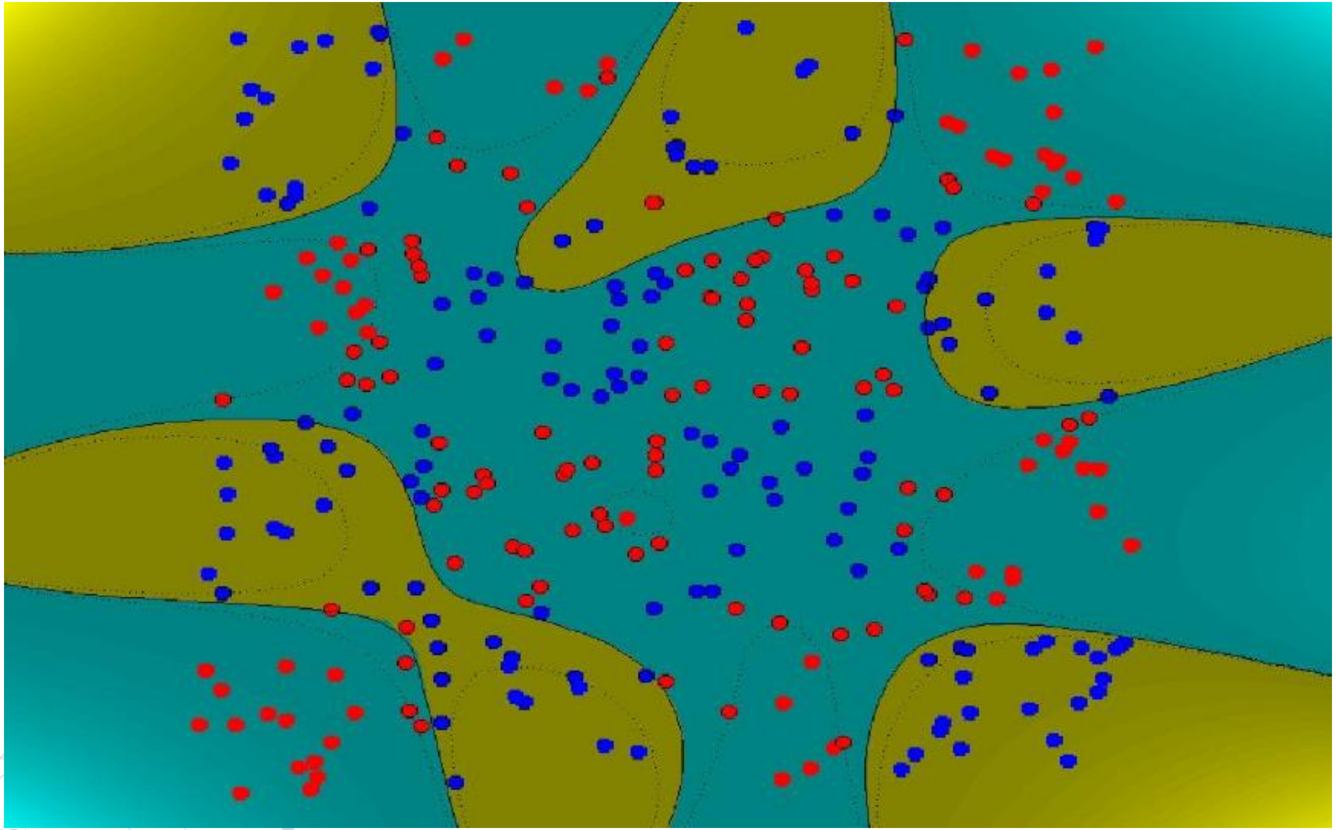
Radial Basis Kernel

ANOTHER DATA SET



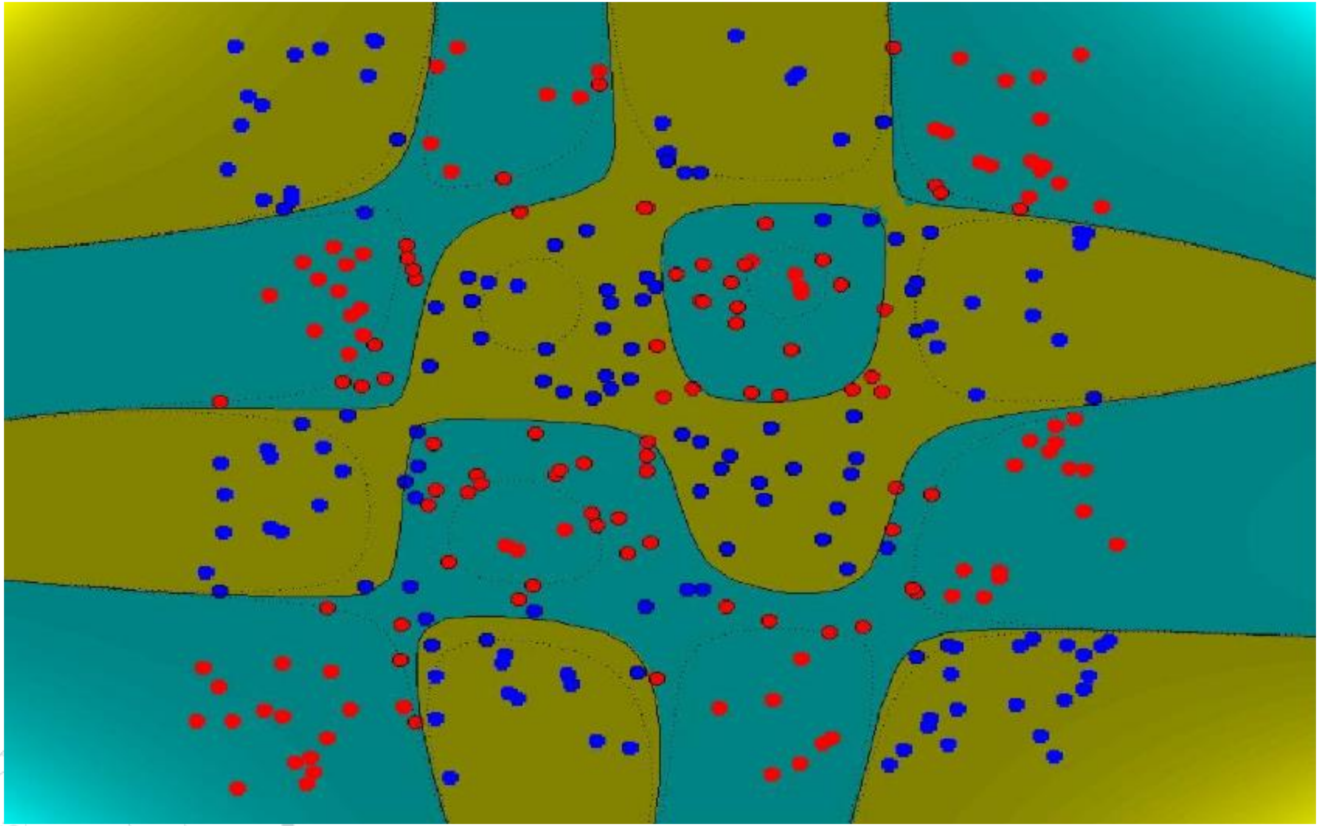
3rd order Polynomial Kernel

ANOTHER DATA SET



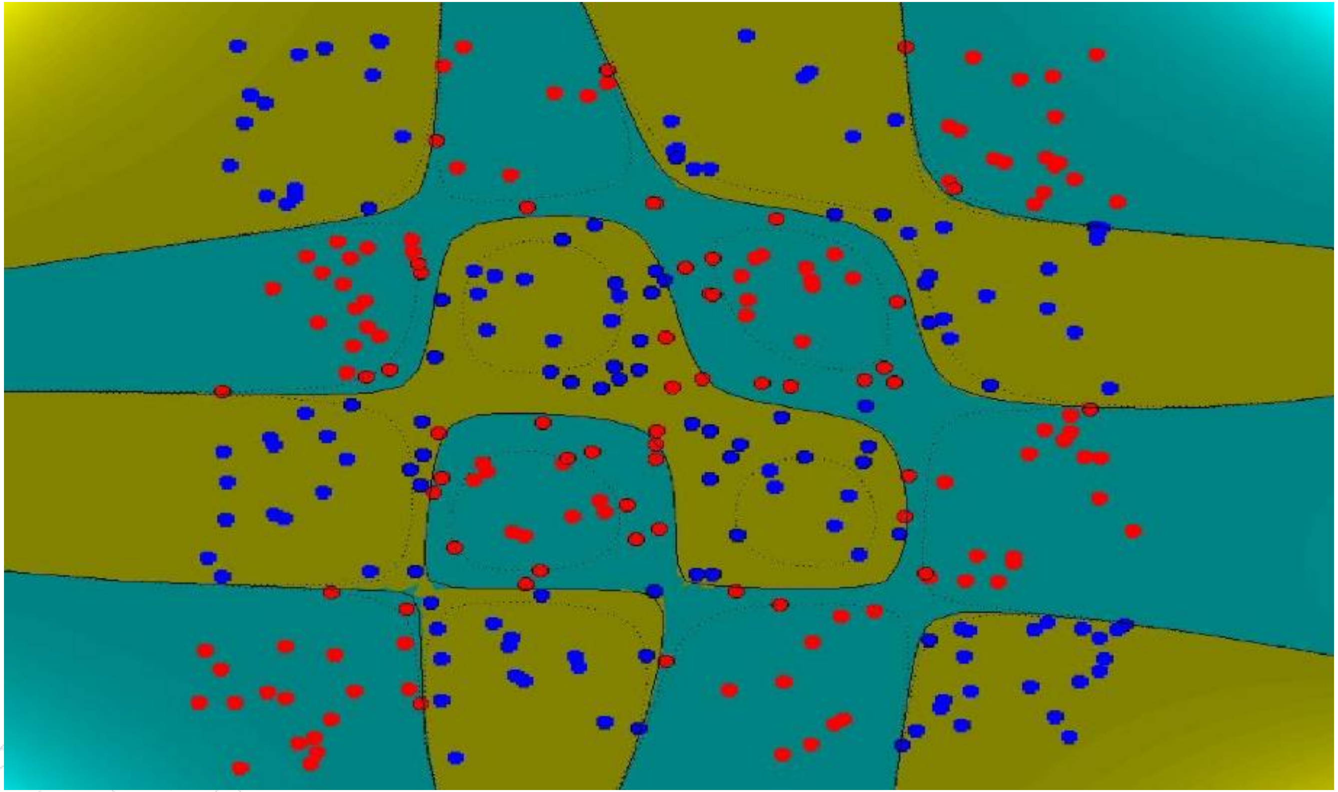
8^{th} order
Polynomial
Kernel

ANOTHER DATA SET



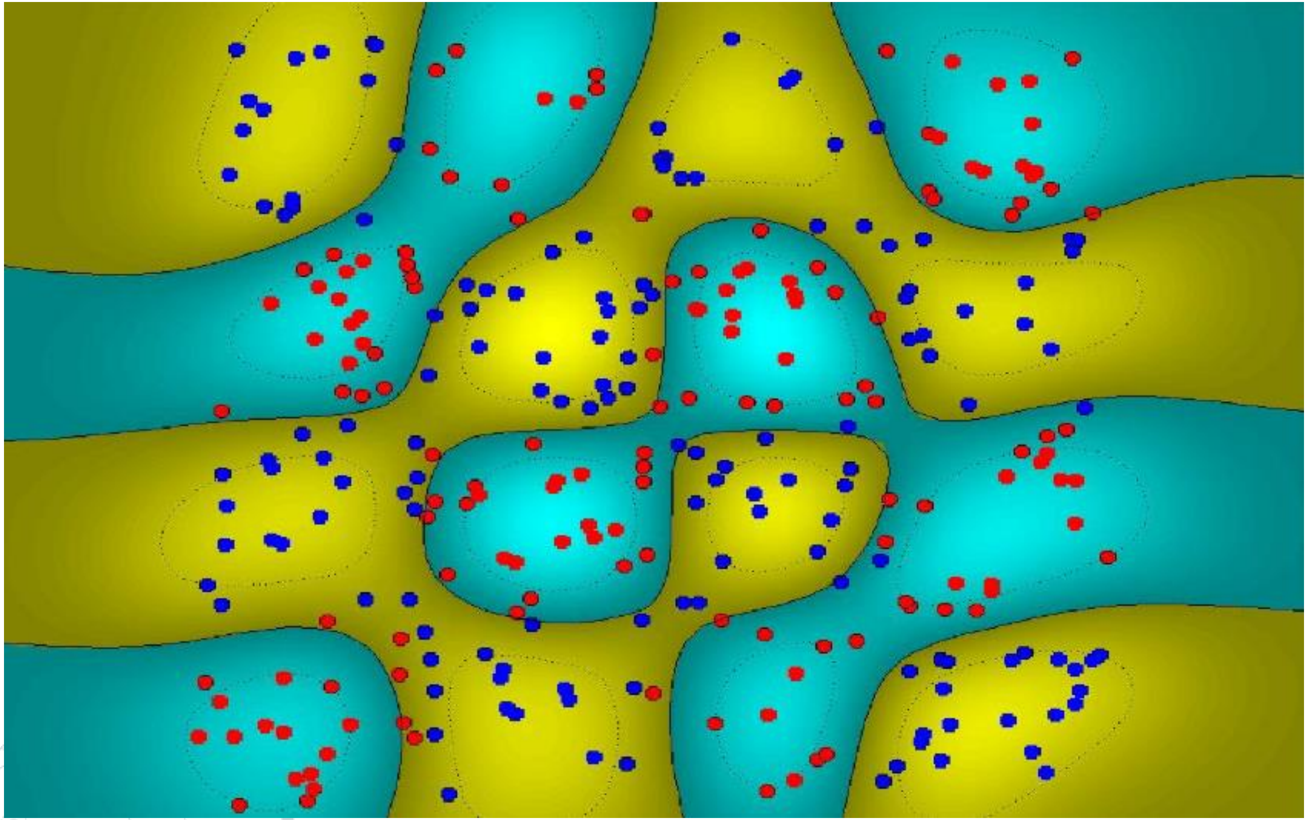
10^{th} order
Polynomial
Kernel

ANOTHER DATA SET



13^{th} order Polynomial Kernel

ANOTHER DATA SET



Radial Basis Kernel

EXTENSIONS: SVM WITH OFFSET

Primal.

minimize $\frac{1}{2}||\theta||^2$

subject to $y(\theta^\top x + \theta_0) \geq 1$ for all data (x, y)

Dual.

maximize $\sum_{(x,y)} \alpha_{x,y} - \frac{1}{2} \sum_{(x,y)} \sum_{(x',y')} \alpha_{x,y} \, \alpha_{x',y'} y y' (x^\top x')$

subject to $\alpha_{x,y} \geq 0$ for all (x, y)

$\sum_{(x,y)} \alpha_{x,y} \, y = 0$

Parameters. $\hat{\theta} = \sum_{(x,y)} \alpha_{x,y} \, y x$

$\hat{\theta}_0 = y - \hat{\theta}^\top x$ where (x, y) is a support vector

EXTENSIONS: SVM WITH ERRORS

Primal.

$$\begin{array}{lll} \text{minimize} & \frac{\lambda}{2}\|\theta\|^2 + \frac{1}{n}\sum_{(x,y)} \xi_{x,y} & \\ \text{subject to} & y(\theta^\top x + \theta_0) \geq 1 - \xi_{x,y} & \text{for all data } (x,y) \\ & \xi_{x,y} \geq 0 & \text{for all data } (x,y) \end{array}$$

Slack variables allow constraints to be violated for a cost.

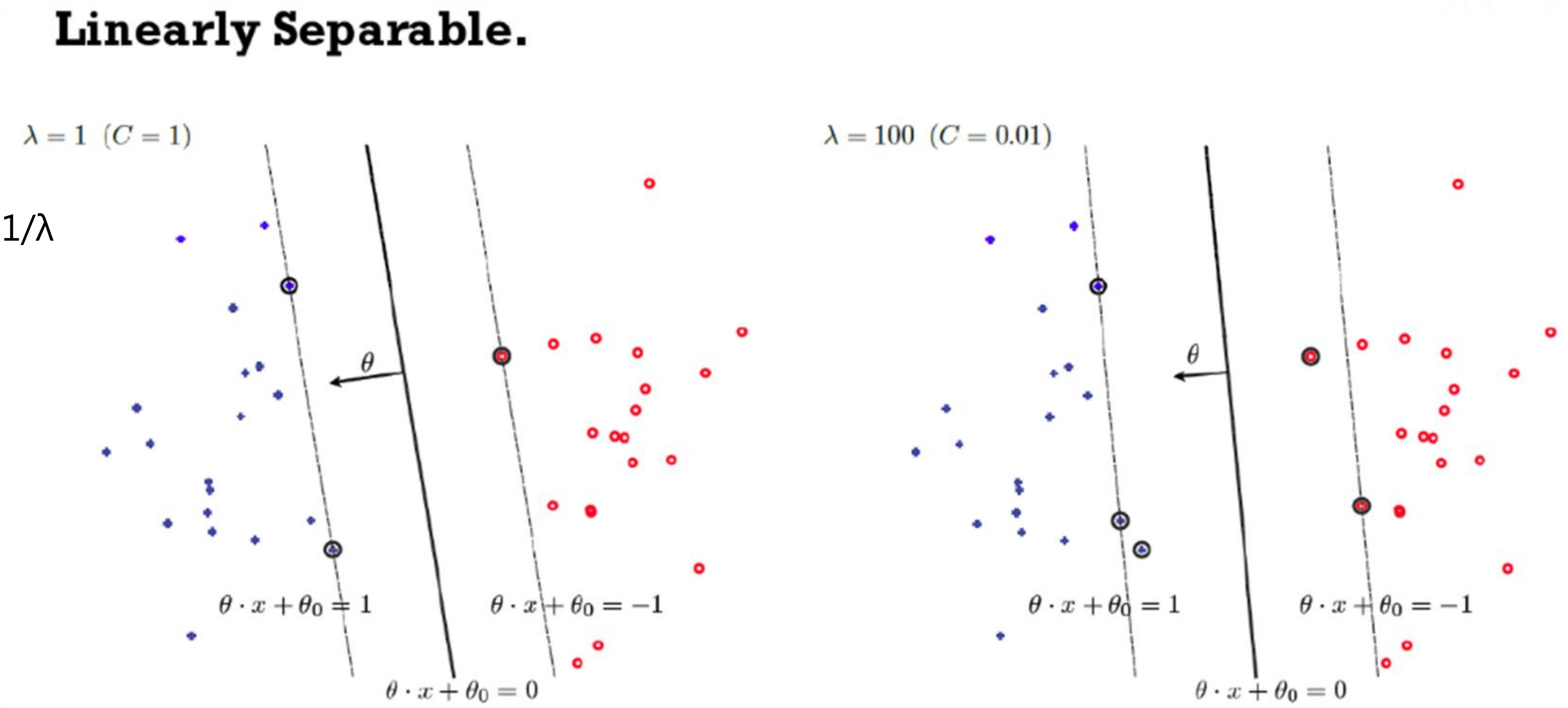
Equivalent Primal.

$$\text{minimize} \quad \frac{\lambda}{2}\|\theta\|^2 + \frac{1}{n}\sum_{(x,y)} \text{Loss}_\text{H}\left(y(\theta^\top x + \theta_0)\right)$$

Hinge-loss classifier with regularization!

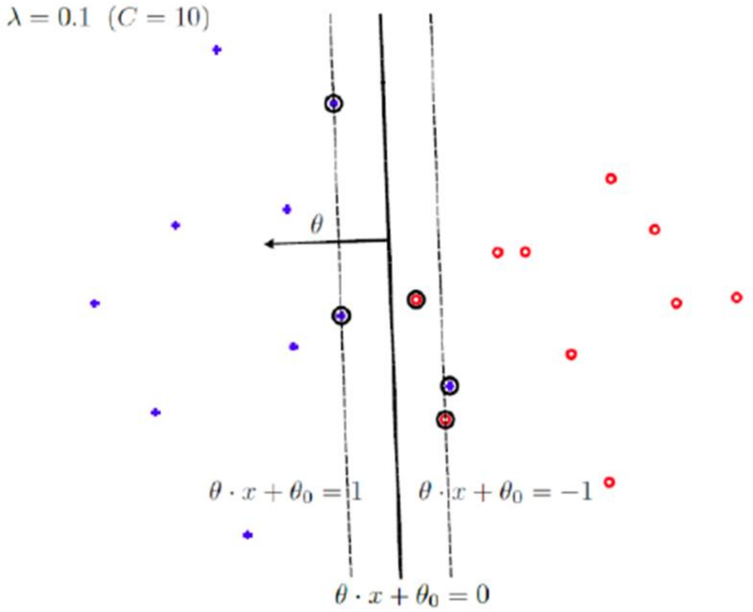
EXTENSIONS: SVM WITH ERRORS

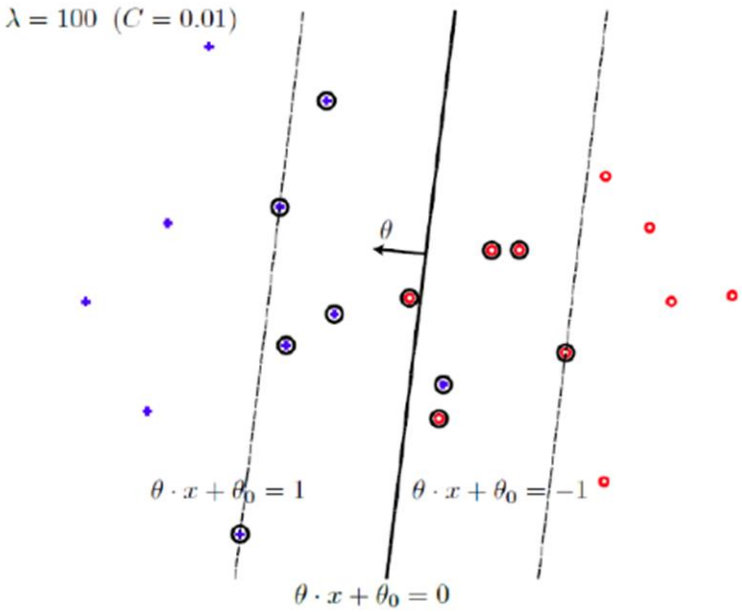
$C = 1/\lambda$



EXTENSIONS: SVM WITH ERRORS

Not Linearly Separable.





EXTENSIONS: SVM WITH ERRORS

Dual.

maximize $\sum_{(x,y)} \alpha_{x,y} - \frac{1}{2}\sum_{(x,y)}\sum_{(x',y')} \alpha_{x,y}\, \alpha_{x',y'} yy'(x^\top x')$

subject to $1/\lambda \geq \alpha_{x,y} \geq 0$ for all (x, y)

$\sum_{(x,y)} \alpha_{x,y}\, y = 0$

Putting limits on what the adversary can do.

There are many efficient solvers for quadratic problems with box constraints.